3.6.1 Technical limitations

- *Limited endurance and range*: Second-generation UAVs were often restricted by fuel efficiency and power limitations. Jet- and piston-engine drones, such as the *RQ-2 Pioneer* or *IAI Scout*, offered endurance between 4 to 8 hours, adequate for tactical missions but insufficient for long-duration strategic reconnaissance.
- *Dependence on line-of-sight communication*: Most drones are operated using radio frequency (RF) communication, which requires a direct line-of-sight connection between the UAV and the ground control station. Terrain obstructions or electromagnetic interference frequently disrupted signals, leading to data loss or mission termination.
- *Limited autonomy*: Although early GPS integration improved navigation, drones still relied heavily on human operators. Autonomous decision-making, obstacle avoidance, and adaptive flight planning were largely absent, limiting operational flexibility.
- *Sensor and payload constraints*: The sensor resolution, stabilization, and data transmission rates were limited by the analog technology of the era. Weather conditions such as cloud cover or wind turbulence often degrade imaging quality, particularly in optical payloads.
- *Navigation and control accuracy*: Early GPS modules lacked precision and robustness against jamming or signal loss. Inertial systems provided some redundancy, but drift errors accumulated over time, affecting positional accuracy during long missions.

3.6.2 Operational and logistical challenges

- *Complex launch and recovery systems*: Drones like the *CL-289* required catapult-assisted launches and parachute recovery, demanding